



7.2.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to explain what was missed in today's lesson.



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can use the ratios of sine and cosine to define the tangent ratio.



7.3.1 Warm-Up: Notice and Wonder



1. What do you wonder?

2. What do you notice?



7.3.2 Exploration: Relating Sine, Cosine and Tangent

1. The table gives the sine and cosine values as decimals for different reference angle measures. Using a calculator and the values in the table, find the ratio of $\frac{\text{sine}}{\text{cosine}}$. Round to the nearest thousandth.

Angle Measure	Sine	Cosine	$\frac{\text{Sine}}{\text{Cosine}}$
10°	0.174	0.985	
20°	0.342	0.940	
30°	0.500	0.866	
40°	0.643	0.766	
50°	0.766	0.643	
60°	0.866	0.500	
70°	0.940	0.342	
80°	0.985	0.174	

2. Use a calculator to find the tangent ($\tan$) of each reference angle.

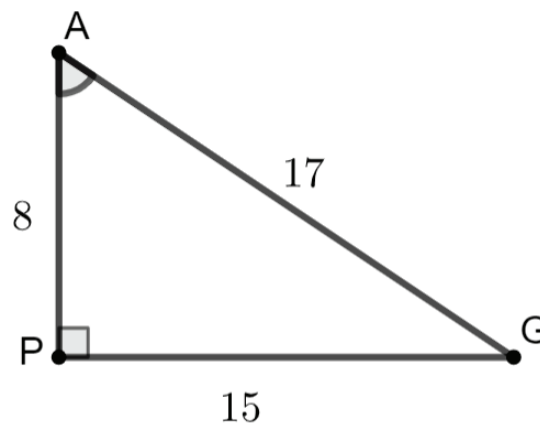
Angle Measure	Tangent
10°	
20°	
30°	
40°	
50°	
60°	
70°	
80°	

3. Write down one notice statement and one wonder statement about the values of $\frac{\text{sine}}{\text{cosine}}$ from the first table and the values of $\tan$ from the second table.

Notice	Wonder



4. $\triangle APG$ is shown, where $m\angle APG = 90^\circ$, $AP = 8$, $PG = 15$, and $AG = 17$.



- a. Using $\angle PAG$ as the reference angle, complete the table.

Sine $\left(\frac{\text{opposite}}{\text{hypotenuse}}\right)$	Cosine $\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right)$	Tangent $\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

- b. Multiply the sine and the cosine ratios from the table.

$$\underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- c. Rewrite sine · cosine using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. If possible, simplify.

- d. Divide the sine ratio by the cosine ratio from the table.

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- e. Rewrite $\frac{\sin}{\cos}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. The resulting ratio is a complex fraction that can be rewritten or simplified. Work with your partner to write the simplified form.

- f. Describe how the ratios of sine and cosine can be used to find the tangent ratio.

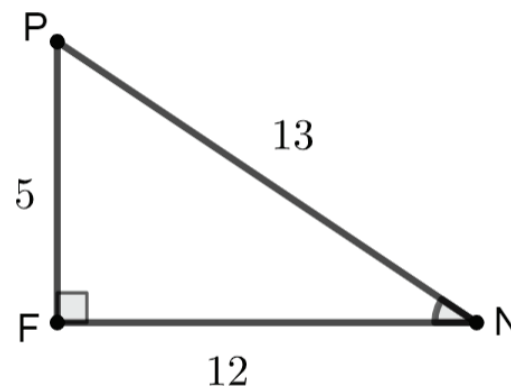


7.3.3 Guided Instruction: Determining Trigonometric Ratios



The tangent ratio of a reference angle can be found using the ratio, $\frac{\sin}{\cos}$.

1. $\triangle PFN$ is shown, where $m\angle PFN = 90^\circ$, $PF = 5$, $FN = 12$, and $PN = 13$.



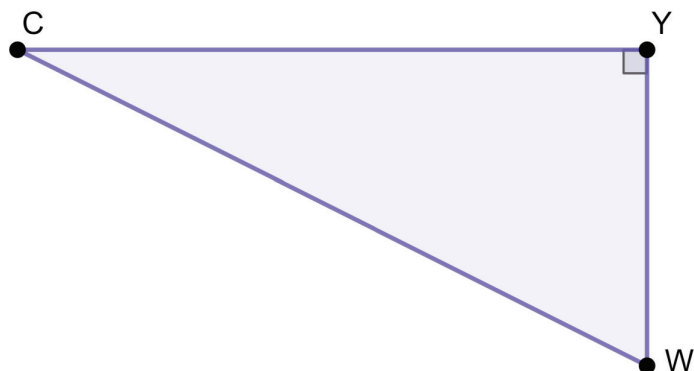
- a. Using $\angle FNP$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

b. Which method, $\frac{\text{sine}}{\text{cosine}}$ or $\frac{\text{opposite}}{\text{adjacent}}$, did you use to find the tangent ratio?

c. Use the method for finding tangent that you did not use to check the ratio.

2. $\triangle CYW$ is shown, where $CY = \sqrt{98}$ and $YW = \sqrt{28}$.



a. Determine the length of $\overline{CW}$.

b. Using $\angle WCY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

c. Using $\angle YWC$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

d. What do you notice about the ratios for $\angle WCY$ and the ratios for $\angle YWC$?



7.3.4 Guided Practice: Trigonometric Ratios

1. $\triangle YNB$, $m\angle NBY = 90^\circ$, $BN = 6\sqrt{2}$, and $YN = 4\sqrt{17}$.

a. Determine the length of $\overline{YB}$.

b. Using $\angle BNY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

2. $\triangle GMK$, $m\angle GKM = 90^\circ$, $GM = 3\sqrt{10}$, and $GK = 3\sqrt{3}$.

a. Determine the length of $\overline{KM}$.

b. Using $\angle KGM$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent



7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.



Unit 7, Lesson 4: Trigonometric Ratios on the Coordinate Plane



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can determine trigonometric ratios using the coordinate plane.